

Proxy dataset creation

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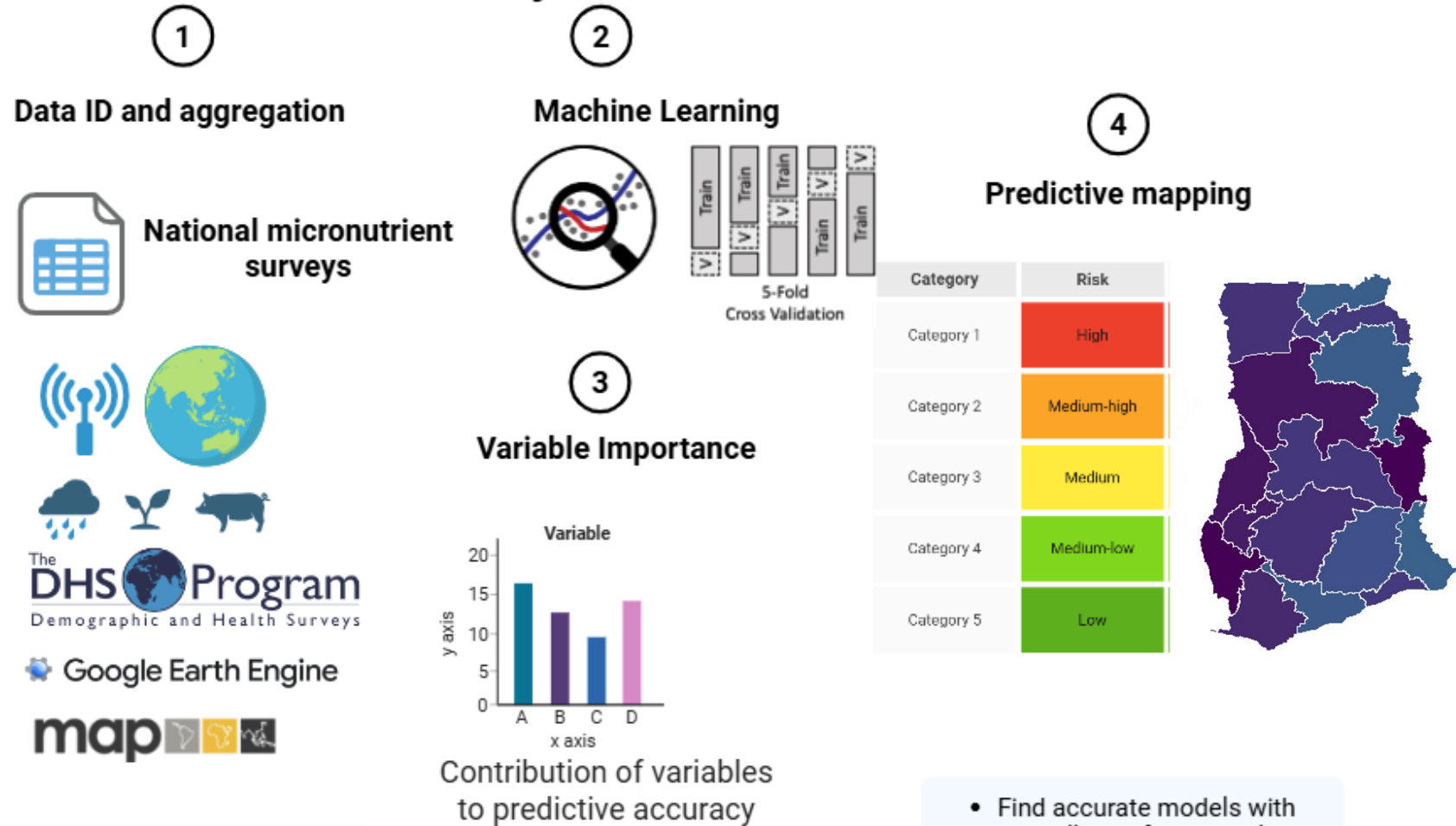


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Goals

1. To generate national and sub-national prevalence estimates for micronutrient deficiency using a wide range of proxy indicators and validated against the prevalence of these same vitamin and mineral deficiencies (VMD) determined by biomarker assessment
2. To identify subnational regions at greatest risk of deficiency
3. To engage with national stakeholders and decision-makers throughout project

Modeling Micronutrient Deficiency Prevalence with Proxy Measures



Merge in time and space:

- Remote sensed climate
- Census and model-based demographics
- Survey-based SES and health
- Modeled burden of disease
- Food pricing and consumption

Contribution of variables to predictive accuracy

- Split merged data into testing and training
- Fit models on training data, and assess accuracy on test data
- Assess which variables contributes most

- Find accurate models with a small set of commonly collected variables
- Assess accuracy in new countries with data
- Map prevalence (with uncertainty) in countries with limited data

Proxy requirements

- Available for every district of every country, surveyed or not, from a non-DUA source
- Independent of the outcome but plausibly related to the outcomes
- Placed at the survey year, or as close to it as the source allows, with the gap recorded.
- Using any type of data: direct survey responses, calculated indices, remote sensed layers, and modeled estimates

Predictor sources: land, climate, agriculture

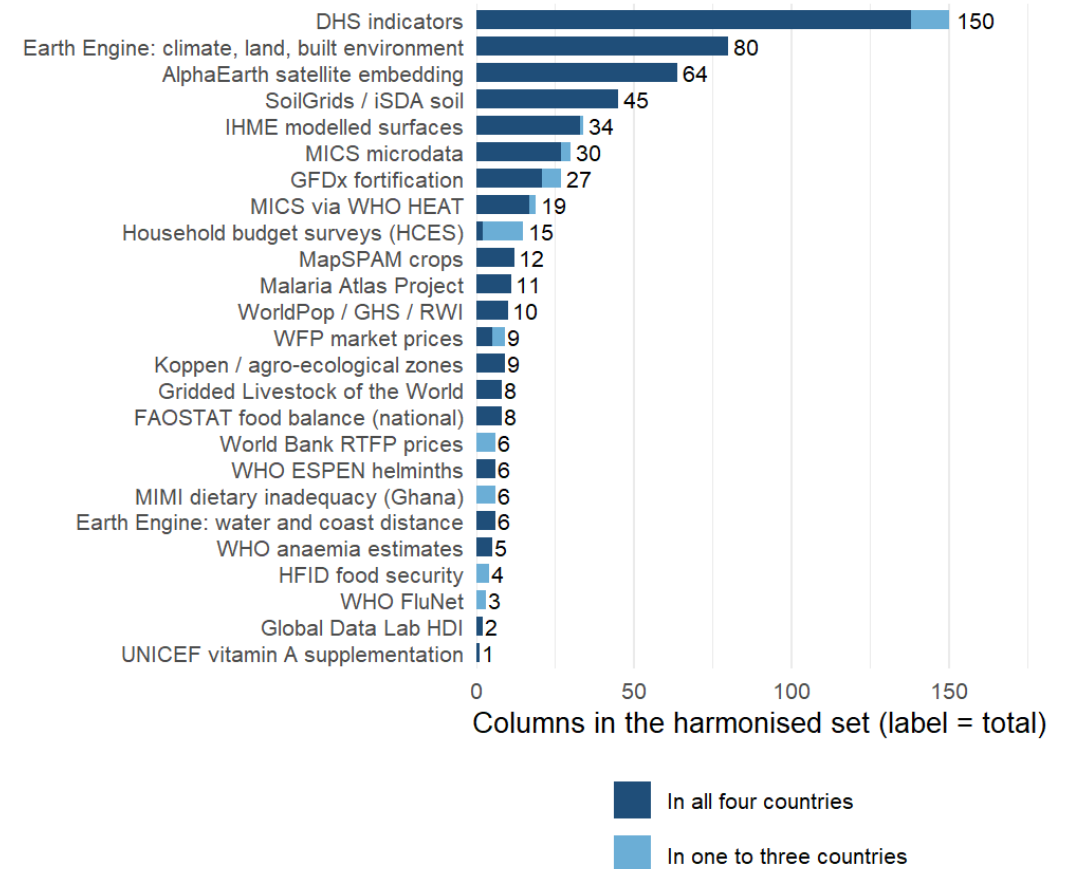
Source	What it is
Earth Engine: TerraClimate	Monthly climate (rain, temperature, evapotranspiration, soil moisture), 4 km; survey year and 1991-2020 normals
Earth Engine: MODIS	Land-surface temperature (1 km), NDVI and EVI (250 m) at the survey year, plus climatology and anomaly
Earth Engine: land cover, lights, built-up	Copernicus land-cover fractions, night lights, GHSL built-up surface, human modification
Earth Engine: productivity	MODIS NPP and GPP, WaPOR evapotranspiration, leaf area, grassland cover
AlphaEarth	Google's 64-number satellite embedding of a year of imagery, 10 m, 2017
SoilGrids / iSDA	Soil nutrients at two depths, pH, texture, organic carbon (iSDA 30 m, SoilGrids 250 m)
Koppen / AEZ	Climate classes and agro-ecological zones: class shares and purity
JRC Surface Water / LSIB	Distance to permanent and seasonal water and to the coast
MapSPAM	Crop area and production by crop, 10 km, 2010
Gridded Livestock of the World	Cattle, sheep, goats, pigs and poultry per km ² , 10 km, 2015
WorldPop / GHSL / GPW / RWI	Population, age-sex structure and density; Meta's relative wealth index

Predictor sources: health, food, households

Source	What it is
IHME modelled surfaces	5 km modelled surfaces: child growth, anaemia, WASH, education, ORS, HIV, malaria, measles vaccination
Malaria Atlas Project	Parasite rate, incidence, mortality, net use and spraying, 5 km, at the survey year
WHO ESPEN	Helminth endemicity and mass-treatment coverage by implementation unit
WHO anaemia; UNICEF VAS; WHO FluNet	National series: anaemia prevalence, vitamin A supplementation, influenza counts
WFP market prices; World Bank RTFP	Monthly market food prices, distance-weighted to districts over the survey year (RTFP: Gambia, Malawi)
FAOSTAT; HFID	National food balance sheets; FEWS NET / IPC food-security phases and indicators
GFDx	Fortification standards and coverage by food vehicle, mostly national
MIMI (Tang et al. 2026)	Modelled dietary micronutrient inadequacy from GLSS7, Ghana regions
DHS	150 district indicators from the paired round, cluster spatial model
MICS microdata; MICS via WHO HEAT	30 indicators from the public microdata (WASH, wealth, education, child feeding, health); 19 regional indicators via WHO HEAT
Household budget surveys (HCES)	Food share, own production, consumption level, 7-day food groups (IHS, GLSS7, IHS4, SLIHS)
Global Data Lab	Subnational human development index

Proxy predictor counts

570 harmonized variables from 28 sources. 510 exist in all four countries and can serve a model for a new country; the other 60 exist in one to three and are used inside the countries that have them.



Types of data

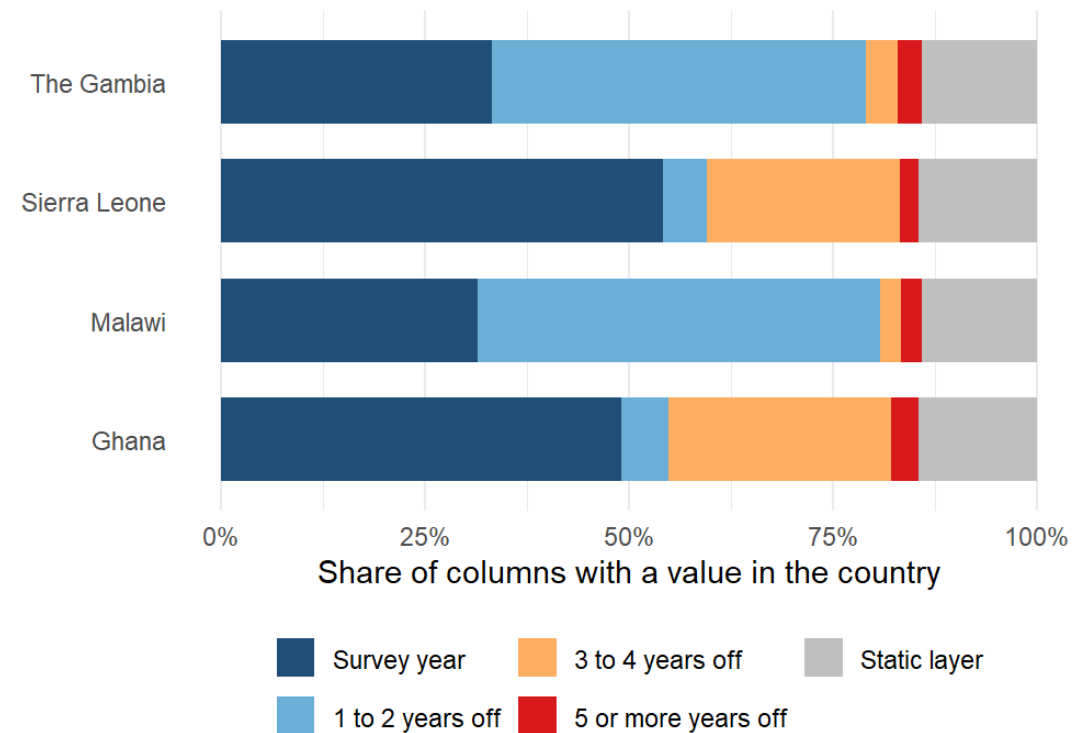
Raw form	Examples	To a district	Time
Gridded raster (30 m to 10 km)	Climate, soil, satellite, IHME and MAP surfaces, population	Zonal mean over the district polygon, area- or population-weighted	Survey year, nearest year, or 30-year normal plus anomaly
Survey with GPS clusters	DHS; MICS 2018 in The Gambia, 2017 in Sierra Leone	DHS: cluster spatial model (BYM2) so every district has a value; MICS: clusters inside the polygon, minimum 30 households	The round paired with the biomarker survey
Survey identified to region or district	MICS in Ghana and Malawi; household budget surveys	Value of the smallest survey unit that contains the district, flagged as the parent's value	The round nearest the survey
National series	FAOSTAT balances, GFDx programme fields, FluNet	Constant within a country; dropped when the model is fitted	Survey year

Linking in space

- 554 districts on GADM 4.1: Gambia 37, Ghana 260, Malawi 243, Sierra Leone 14.
- Every unit carries its GADM code, HASC code and the OCHA p-code, so a join is by code where the source has one.
- Names are matched only as a last resort, through a reviewed alias table with a distance cap, and always on the region-district pair
- Ghana: 260 districts on the 2019 boundaries; sources on the 2012 boundaries (216) or the ten old regions
- Malawi: 243 Traditional Authorities; sources at district level (28) are broadcast down and flagged.
- The Gambia: 37 districts
- Sierra Leone: 14 districts.

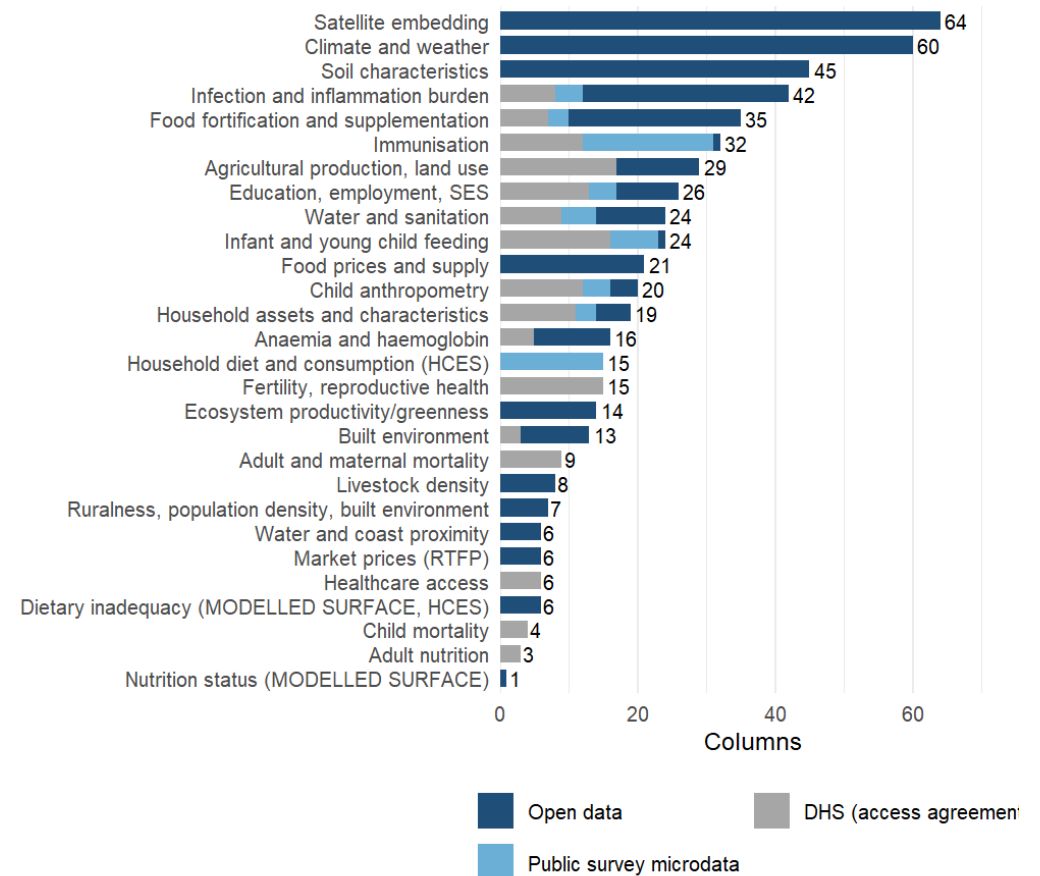
Linking in time

- Surveys: The Gambia 2018, Ghana 2017, Malawi 2016, Sierra Leone 2013.
- 131 variables are from the survey year in every country and 78 are static layers.
- 306 are three or more years off in at least one country:
 - the DHS round paired with Ghana's 2017 survey is from 2014
 - Sierra Leone's 2013 survey predates the satellite embedding and its nearest MICS by four years
 - MapSPAM's 2010 crop map is the oldest layer everywhere.



Variable domains

- 28 domains.
- Each domain enters our best model as its leading principal components, to 80% of variance and at most twelve, learned on the training districts only.



Proxy dataset CONSORT

Step	What happens	Columns
Assemble	Base extraction on the spine, food environment, extra sources, add-back sources	
Append blocks	Household budget surveys, MICS microdata, RTFP prices, MODIS NDVI, climate normals	614
Exclude	Rules with a written reason: data defects, superseded layers, redundant twins	570
Stamp	Access tier, within-country variation, alignment rule, year used and gap, modelled-surface flag	570
Fit-time policy	Drop national constants and leakage columns	525
Transport matrix	Per country: coverage of at least 70% and some variation; then the four-country intersection	464

Ensemble Machine Learning

Competition” of algorithms

- Combinations of simple regressions and more complex machine learning models to allow interactions and nonlinear associations.

Best “team” wins

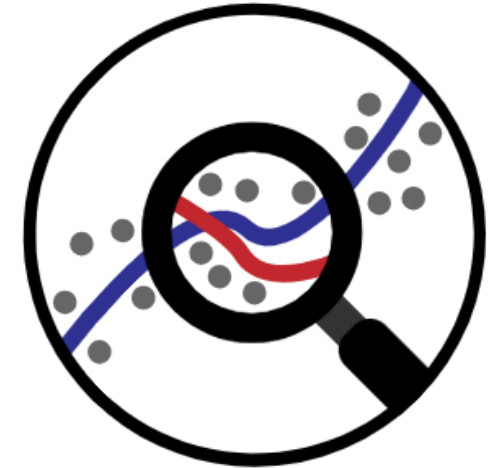
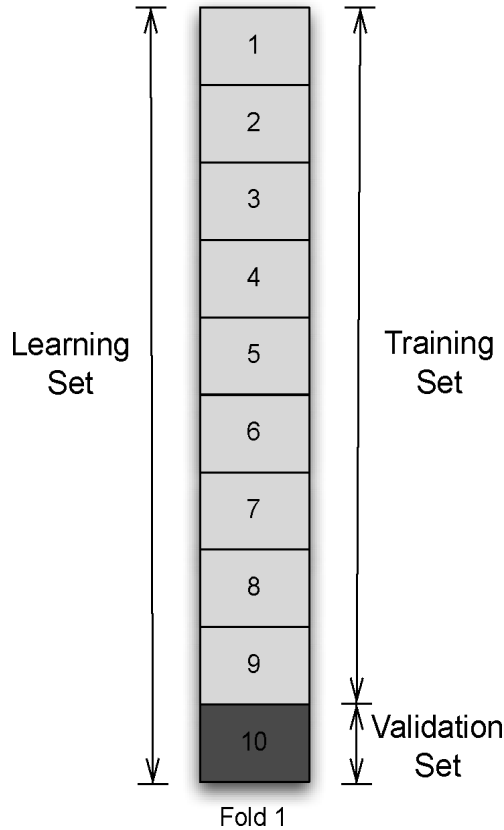
- Weighted combination of algorithms

Performance judged on independent data

- V-fold cross validation (Internal data splits) to avoid overfitting

Library:

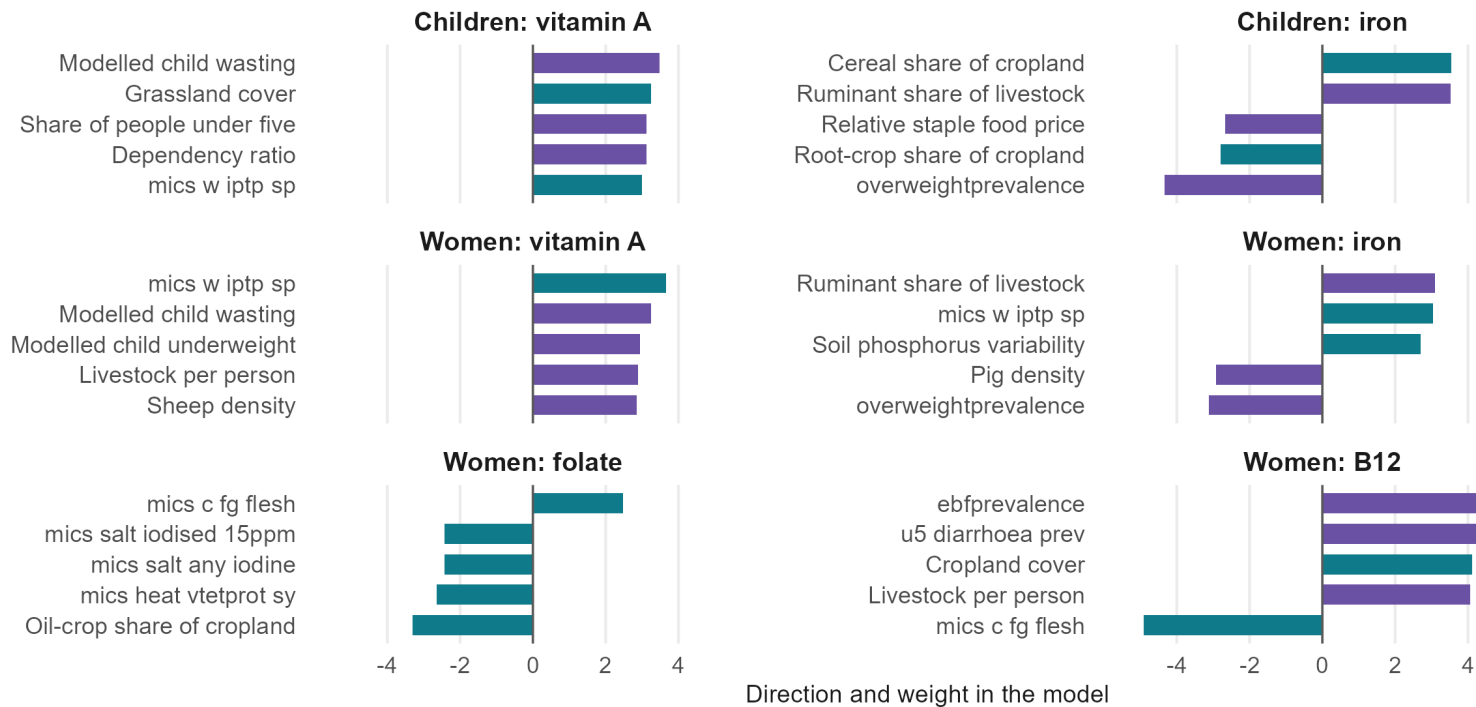
- Simple mean
- Penalized regressions
- Random forest models
- Gradient Boosting models
- PCA-HAL
- Bayesian Additive and Regression trees
- Bayesian Geostatistical Models



Variable importance

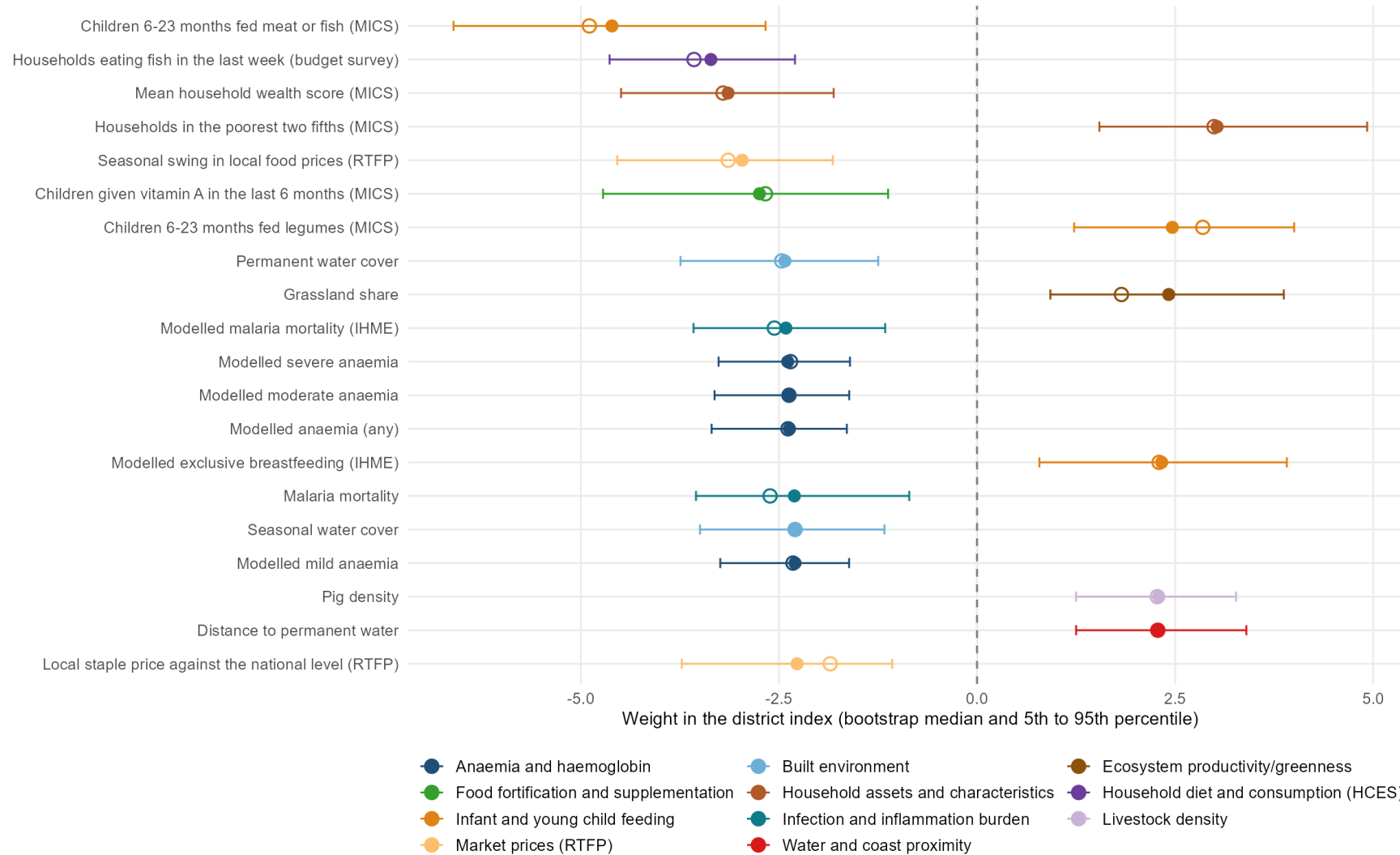
Bars to the right mean the district ranks worse. Colour = the kind of data.

■ Modelled or administrative ■ Remotely sensed environment

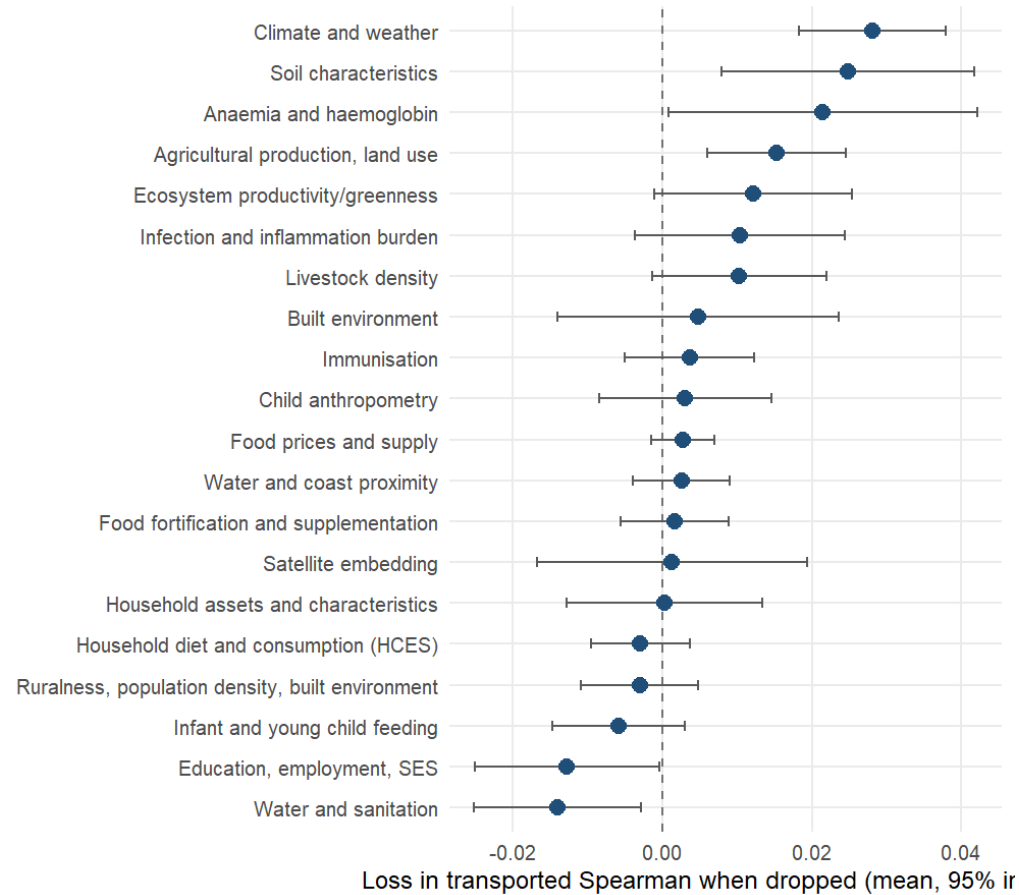


These weights say where deficiency is. They are not causes and not things to change.

Variable importance: Malawi B12 concentrations in women



Variable importance- cross country models



What works in the cross-country transport models

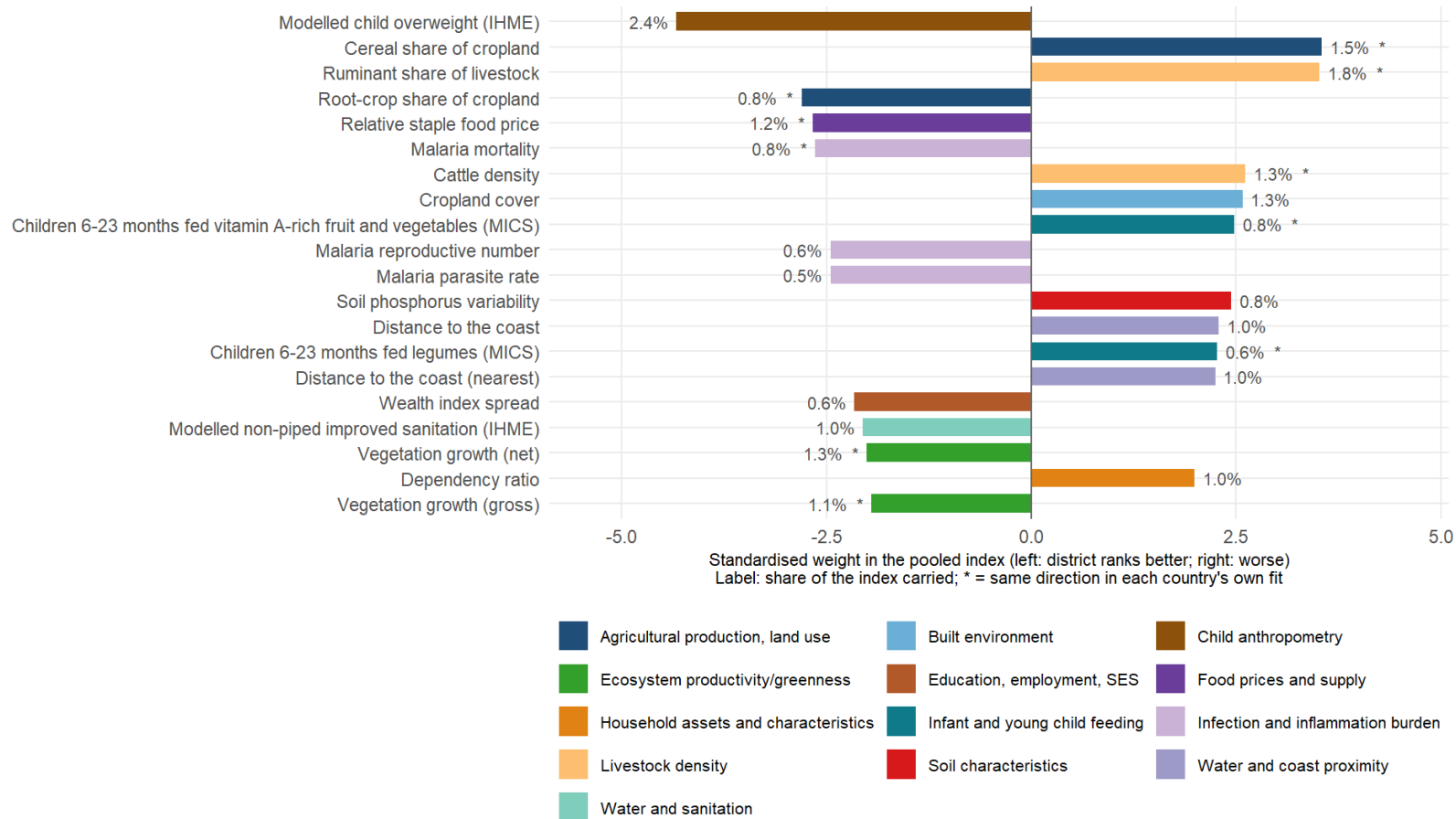
- The landscape gradient: grassland and cropland cover, the share of ruminants in the herd (top twenty in 14 of the 22 held-out fits), cattle and sheep density, cereal-heavy cropping, and low plant productivity. Dry, grassy, pastoral, unproductive districts rank worse on every outcome.
- The modelled child-growth and anaemia surfaces: wasting, underweight and anaemia point to worse status; overweight points the other way and behaves as a wealth proxy.
- The food economy: the local staple price relative to the national level appears in 13 fits across 5 outcomes.
- Remoteness and demography: distance to the coast and the dependency ratio, in about a third of the fits.
- Malaria intensity recurs, with the sign a nutritional story would not predict: the high-malaria zones are the wetter, more productive ones. Single household-survey items rarely recur; malaria treatment in pregnancy and infant feeding of legumes are the exceptions.

Appendix

Sources and time-alignment

Source	Columns	Tier	Domains	Year rule	Largest gap (years)
DHS indicators	150	DHS (access agreement)	16	dhs_round	3
Earth Engine: climate, land, built environment	80	Open data	4	access_2019, ccnl_2013, clim_anomaly, clim_normals, gee_t...	6
AlphaEarth satellite embedding	64	Open data	1	alphaearth_2017	4
SoilGrids / iSDA soil	45	Open data	1	soil_static	static
IHME modelled surfaces	34	Open data	7	ihme_to_2017, ihme_to_2018, ihme_to_2019	1
MICS microdata	30	Public survey microdata	7	mics_microdata	4
GFDx fortification	27	Open data	3	gfdx_ntd_2015, gfdx_survey_year, gfdx_who_anaemia_2011, g...	6
MICS via WHO HEAT	19	Public survey microdata	1	mics_heat_round	3
Household budget surveys (HCES)	15	Public survey microdata	1	hces_year	5
MapSPAM crops	12	Open data	1	spam_2010	8
Malaria Atlas Project	11	Open data	1	map_blood_2012, map_survey_year	6
WorldPop / GHS / RWI	10	Open data	3	ghsl_epoch, rwi_2019, wpop_2020, wpop_density_sy	7
Koppen / agro-ecological zones	9	Open data	1	aez_static, koppen_static	static
WFP market prices	9	Open data	1	fprice_window	0
FAOSTAT food balance (national)	8	Open data	1	fao_supply	0
Gridded Livestock of the World	8	Open data	1	glw4	3
Earth Engine: water and coast distance	6	Open data	1	water_static	static
MIMI dietary inadequacy (Ghana)	6	Open data	1	mimi_glss7	1
WHO ESPEN helminths	6	Open data	1	espen_mda, espen_nearest	3

Variable importance for child iron.



Predictor set by domain and tier

Domain	Columns	Open	Public microdata	DHS	Vary within country	Modelled surfaces
Satellite embedding	64	64	0	0	64	0
Climate and weather	60	60	0	0	60	0
Soil characteristics	45	45	0	0	45	0
Infection and inflammation burden	42	30	4	8	39	10
Food fortification and supplementation	35	25	3	7	10	0
Immunisation	32	1	19	12	32	1
Agricultural production, land use	29	12	0	17	29	0
Education, employment, SES	26	9	4	13	26	4
Infant and young child feeding	24	1	7	16	24	1
Water and sanitation	24	10	5	9	24	10
Food prices and supply	21	21	0	0	12	0
Child anthropometry	20	4	4	12	20	4
Household assets and characteristics	19	5	3	11	19	0
Anaemia and haemoglobin	16	11	0	5	9	11
Fertility, reproductive health	15	0	0	15	15	0
Household diet and consumption (HCES)	15	0	15	0	15	0
Ecosystem productivity/greenness	14	14	0	0	14	0
Built environment	13	10	0	3	13	0
Adult and maternal mortality	9	0	0	9	9	0
Livestock density	8	8	0	0	8	0
Food and nutrition security (FNS)	5	5	0	0	5	0